

JavaScript Short Questions

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1. Variables & Scoping

What is the difference between var, let, and const in JavaScript?

- var is function-scoped and can be reassigned.
- let is block-scoped and can be reassigned.
- const is block-scoped and cannot be reassigned after initialization.

2. Operators

What is the difference between == and ===? Give one example.

- == compares values after type conversion.
- === compares both value and data type.

Example:

```
1      5 == "5" // true
2      5 === "5" // false
```

3. Loops

What is the difference between a for loop and a for-of loop?

- A for loop uses an index and condition to control iteration.
- A for...of loop iterates directly over the values of an iterable such as an array.

Example:

```
1      for (let i = 0; i < arr.length; i++) {
2          console.log(arr[i]);
3      }
4
5      for (const item of arr) {
6          console.log(item);
7      }
```

4. Functions

What is an arrow function? Write a simple example.

An arrow function is a shorter syntax for writing functions.

Example:

```
1    const addOne = x => x + 1;
```

5. Advanced Functions

What are an IIFE, a recursive function, and a generator function? Write one key characteristic of each.

IIFE

An Immediately Invoked Function Expression executes immediately after it is defined.

```
1    (function () {  
2        console.log("Hello");  
3    })();
```

Recursive Function

A recursive function calls itself and requires a base case.

```
1    function countdown(n) {  
2        if (n === 0) return;  
3        countdown(n - 1);  
4    }
```

Generator Function

A generator function uses yield to pause execution and return values.

```
1    function* numbers() {  
2        yield 1;  
3        yield 2;  
4    }
```

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